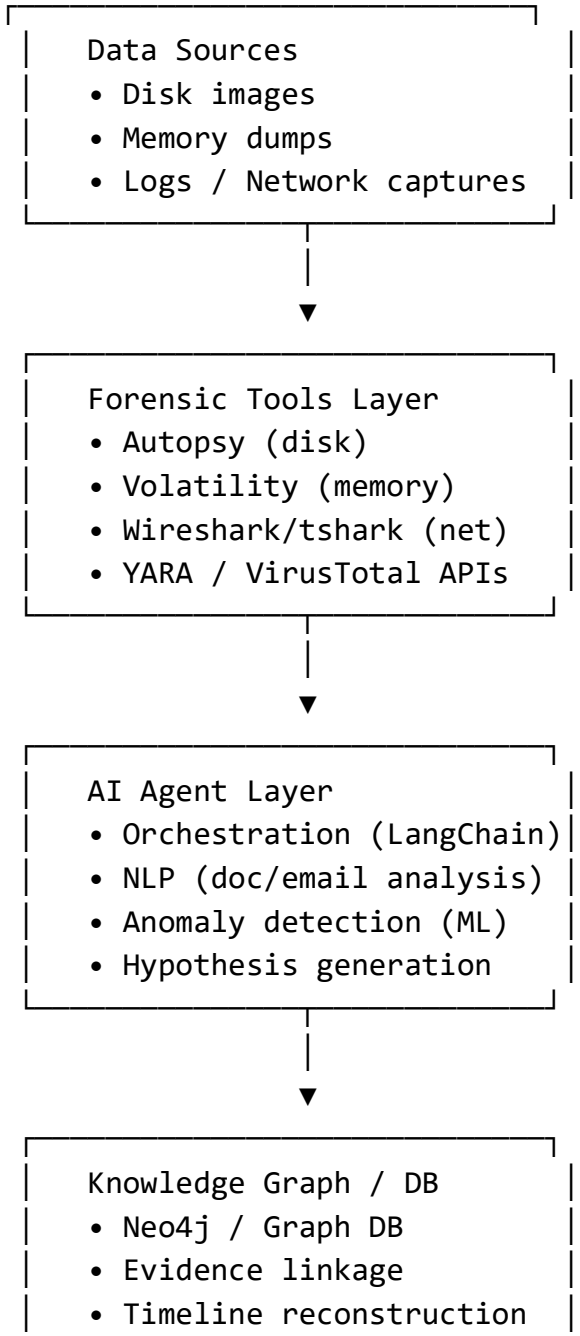
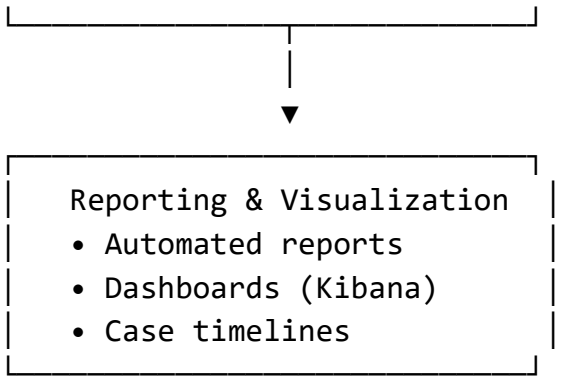
Perfect — let's structure this step by step so you have a clear **reference architecture**, anomaly-handling strategy, tool extensibility, and a **five-day pipeline plan** for your proof-of-concept.

🏗️ Reference Architecture Diagram (Conceptual)

Code





🔎 Handling Anomalies

To “add up” anomalies for your use case:

- **Define anomaly categories**:
  - File system anomalies (hidden files, deleted logs).
  - Memory anomalies (suspicious processes, injected code).
  - Network anomalies (unusual ports, exfiltration patterns).
- **Scoring system**: Assign weights to anomalies (e.g., hidden partition = high severity, odd timestamp = medium).
- **Aggregate anomalies**:
  - Use a **rule-based engine** or ML classifier to combine anomalies into a **risk score**.
  - Example: If multiple anomalies cluster around the same user or time window, flag as “critical event.”
- **AI role**: Summarize anomalies into human-readable insights (“User X deleted logs after accessing sensitive file Y”).

🔧 Keeping It Open for Other Tools

Yes — design the pipeline modularly:

- Use **adapters/wrappers** for each forensic tool (Autopsy, Volatility, Wireshark).
- AI agent orchestrates tasks via APIs or CLI calls.
- Future tools (e.g., Sleuth Kit, Plaso, ElasticSearch) can be plugged in without redesigning the whole system.

- Think of it as a **“tool registry”** the agent can call depending on the evidence type.

🗓️ Five-Day Pipeline Plan

Day 1 – Setup & Architecture

- Install Autopsy, Volatility, Wireshark.
- Define modular architecture (data ingestion → AI agent → reporting).
- Prepare sample forensic dataset (disk image, logs).

Day 2 – Disk Forensics Integration

- Run Autopsy on sample disk image.
- AI agent parses Autopsy output (files, metadata).
- Implement anomaly scoring for file system artifacts.

Day 3 – Memory & Network Forensics

- Integrate Volatility for memory dumps.
- Add Wireshark/tshark for network captures.
- AI agent aggregates anomalies across disk, memory, and network.

Day 4 – Knowledge Graph & Reporting

- Build Neo4j graph linking evidence.
- AI agent generates timeline + hypotheses.
- Create automated reporting templates.

Day 5 – Presentation-Ready Solution

- Run full pipeline on test dataset.
- Produce final report with anomalies, timeline, and AI insights.
- Demo dashboard + architecture diagram.

✅ By Day 5, you’ll have a **running solution** that demonstrates AI-driven orchestration of forensic tools, anomaly aggregation, and automated reporting — perfect for presentation.

Would you like me to also **draft a sample anomaly scoring matrix** (with weights and categories) so you can plug it directly into your proof-of-concept?